Experiment 6

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Branch: BE CSE **Section/Group: KRG 3A**
Semester: 6th **Date of Performance: 02/03/26**
Subject Name: Full Stack Development **Subject Code: 23CSH-309**

Aim:
To design and implement a Spring Boot REST API using Layered Architecture, applying Inversion of Control (IoC) and Dependency Injection (DI) to build a structured backend for the HealthHub application, including CRUD operations, DTO mapping, validation, and global exception handling.

Objective:
After completing this experiment, students will be able to:

1. Understand Spring Boot fundamentals
   - Learn the role of Spring Boot in backend development.
   - Configure and run a basic Spring Boot application.

2. Apply IoC and Dependency Injection - Understand how Spring manages object creation and dependencies.
   - Use annotations like @Autowired, @Service, and @Repository.

3. Implement Layered Architecture
   - Structure the application using Controller, Service, and Repository layers.
   - Understand separation of concerns in backend design.

4. Develop RESTful APIs

- Create CRUD operations using Spring Boot REST controllers.
- Design APIs using proper HTTP methods (GET, POST, PUT, DELETE).

5. Use DTOs for API communication - Implement DTO mapping to separate internal models from API responses.

6. Implement Validation
   - Apply validation using Jakarta Validation annotations (@NotNull, @Email, etc.).

7. Handle Exceptions Globally
   - Implement global exception handling using @ControllerAdvice.

8. Build the HealthHub Backend Module
   - Develop backend services for patient/health record management.

<u>Implementation/Code</u>:

PatientController.java:

```
package com.healthhub.controller;

import java.util.List;

import
org.springframework.beans.factory.annotation.A
utowired;
import
org.springframework.web.bind.annotation.*;

import com.healthhub.model.Patient;
import com.healthhub.service.PatientService;
```

```
@RestController
@RequestMapping("/patients")
public class PatientController {

    @Autowired
    private PatientService service;

    @PostMapping
    public Patient createPatient(@RequestBody
Patient patient) {
        return service.createPatient(patient);
    }

    @GetMapping
    public List<Patient> getAllPatients() {
        return service.getAllPatients();
    }

    @GetMapping("/{id}")
    public Patient getPatient(@PathVariable Long
id) {
        return service.getPatient(id);
    }

    @PutMapping("/{id}")
    public Patient updatePatient(@PathVariable
Long id, @RequestBody Patient patient) {
        return service.updatePatient(id, patient);
    }

    @DeleteMapping("/{id}")
    public void deletePatient(@PathVariable
Long id) {
        service.deletePatient(id);
    }
}
```

PatientRepository.java:

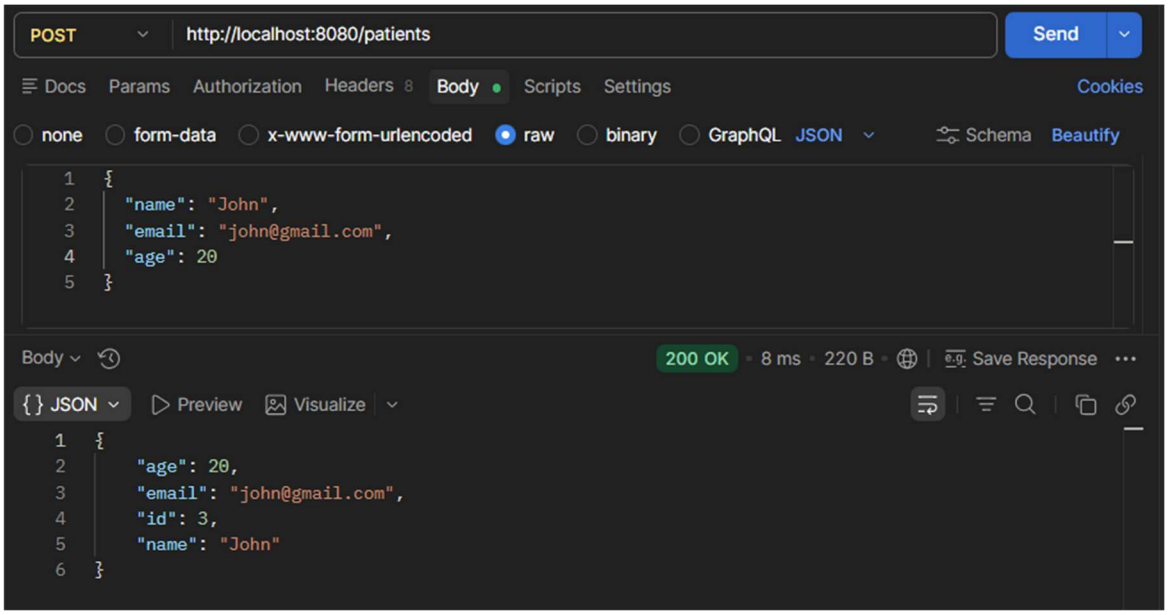
```
package com.healthhub.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.healthhub.model.Patient;

public interface PatientRepository extends JpaRepository<Patient, Long> {

}
```

PatientService.java:

```
package com.healthhub.service;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.healthhub.model.Patient;
import com.healthhub.repository.PatientRepository;

@Service
public class PatientService {

    @Autowired
    private PatientRepository repository;

    public Patient createPatient(Patient patient) {
        return repository.save(patient);
    }

    public List<Patient> getAllPatients() {
        return repository.findAll();
    }

    public Patient getPatient(Long id) {
```

```
        return repository.findById(id)
                .orElseThrow(() -> new RuntimeException("Patient not found"));
    }

    public Patient updatePatient(Long id, Patient newPatient) {

        Patient patient = getPatient(id);

        patient.setName(newPatient.getName());
        patient.setEmail(newPatient.getEmail());
        patient.setAge(newPatient.getAge());

        return repository.save(patient);
    }

    public void deletePatient(Long id) {
        repository.deleteById(id);
    }
}
```

Output:



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GET http://localhost:8080/patients Send

Docs Params Authorization Headers 6 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

Body 200 OK 8 ms 284 B Save Response

{ } JSON Preview Visualize


```
[
    {
        "age": 20,
        "email": "kaur@gmail.com",
        "id": 1,
        "name": "Harleen"
    },
    {
        "age": 29,
        "email": "kavin@gmail.com",
        "id": 2,
        "name": "Kavin"
    }
]
```


PUT http://localhost:8080/patients/1 Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

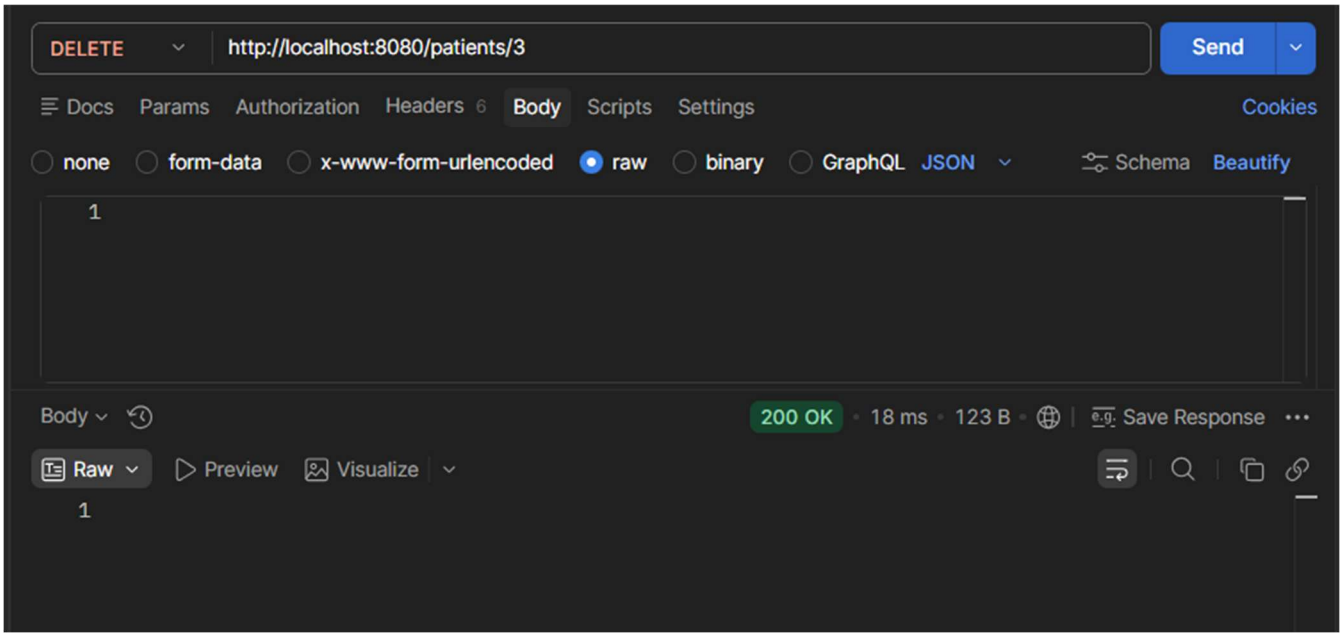
none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{
  "name": "Harleen",
  "email": "kaur@gmail.com",
  "age": 20
}
```


Body 200 OK 42 ms 223 B Save Response

{ } JSON Preview Visualize


```
{
    "age": 20,
    "email": "kaur@gmail.com",
    "id": 1,
    "name": "Harleen"
}
```



Learning Outcomes:

- **Develop a Spring Boot application**
  Create and run a basic Spring Boot project and understand its role in modern backend development.
- **Apply Inversion of Control (IoC) and Dependency Injection (DI)**
  Implement IoC and DI concepts using Spring annotations such as @Autowired, @Service, and @Repository to manage dependencies efficiently.
- **Design applications using Layered Architecture**
  Structure backend applications into Controller, Service, and Repository layers, ensuring proper separation of concerns and maintainable code.
- **Create RESTful APIs with CRUD functionality**
  Develop REST APIs using appropriate HTTP methods (GET, POST, PUT, DELETE) to perform Create, Read, Update, and Delete operations.
- **Implement DTO-based data transfer**
  Use Data Transfer Objects (DTOs) to separate internal entity models from API request and response structures.